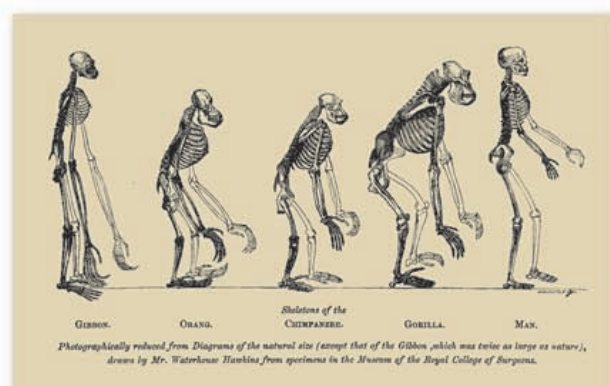


“Man with all his noble qualities... still bears in his bodily frame the indelible stamp of his lowly origin.”

CHARLES DARWIN, FROM *THE DESCENT OF MAN*, 1871

Origin of Species

The theory of evolution is often called “Darwinism,” after the British naturalist Charles Darwin. In 1859 Darwin published the results of his research in the first evolutionary theory based on scientific evidence. His controversial work known as *On the Origin of Species* changed our view of the world.



Evolving forms

T. H. Huxley (see pp.342–43) was the foremost exponent of Darwin's theories. This frontispiece to his *Evidence as to Man's Place in Nature*, 1863, reveals the similarities in the skeletons of (from left to right) gibbon, orangutan, chimpanzee, gorilla, and human.

time, contradicting the Biblical notion that they had been unchanged since

their creation by God. Chambers' ideas were unscientific and unsubstantiated, but the very idea of evolution was enough to ensure that the book sold in huge numbers.

Natural selection

From 1831 to 1836, Darwin traveled as a naturalist aboard HMS *Beagle*, on a scientific survey of South American waters. During the voyage he found evidence to support the geologist Charles Lyell's idea that the world had evolved through gradual processes (see BEFORE), but it was not until several years later that Darwin began to forge his own theory. In the meantime he read Thomas Malthus's 1798 “Essay on the Principle of Population” (see p.291), which argued that the size of an animal population is limited by its food supply. When Darwin's friend, the ornithologist John Gould, realized that the species of finches that Darwin had brought back from his voyage shared common ancestors, Darwin wondered why. He concluded that those animals most suited to acquiring food would survive and pass on their characteristics to their offspring, while the unsuited would die out—a concept that Darwin called “natural selection,” but British philosopher Herbert Spencer later called “the survival of the fittest.”

Although the uproar caused by *On the Origin of Species by Means of Natural Selection* might suggest otherwise, the idea that humankind and animals had evolved from more primitive creatures was nothing new, even in Darwin's day. The first theory that all living creatures are descended from shellfish was proposed as early as the 6th century BCE. Some 2,500 years later, in 1844, Scottish encyclopedist Robert Chambers scandalized Victorian society by anonymously publishing *Vestiges of the Natural History of Creation*. Chambers argued that fossil evidence suggested that animal species had changed over



Darwin's finches

Darwin's illustrations of some of the species of finches he discovered in the Galápagos Islands. The realization that the separate species were all derived from common ancestors influenced Darwin's theory of natural selection.

BRITISH NATURALIST (1809–82)